COMP 3500SEF · Final Report

Warehouse Logistics Management System
Group 42: Tang Yi Jie, Yin Pei Xu, Tang Chao, Yang Ting Zhang, Wu Peng Bo, Chung Ngon Man, Li Mu Qiu.

1. Scope

The goals are traceable order decisions (state changes produce notifications; confirmation writes a ledger entry and a notification) and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application uses Next.js, TypeScript, Prisma and SQLite. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only; data integrity is enforced by transactional guards (§5).

2. Development process

Increments cover product and specification changes and recorded process milestones, each with a trigger and verified outcome.

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selection	Responsibilities and contracts recorded (increment register)
17 Oct 2026 test-led review	Assess order and stock functions	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Failed confirmation endpoint and missing stock ledger (increment register)	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Foundation codebase received	Customer, staff and admin workflows verified in final regression (browser checks)
Driver and privacy, 2 Nov 2026	Delivery and data minimisation	Driver assignment and data-minimisation checks pass (tests)
Documentation hardening, 10–11 Nov 2026	Model and evidence review	Diagrams and traceability aligned (traceability)
Export simplification, 11 Nov 2026	11 Nov automated walkthrough (issue #15)	Selected-date CSV export in three actions; regression passed
Availability tests, 11–12 Nov 2026	Regression gap in integration-health checks	Suite increased to 38 (unit.log)

3. Requirements

Functional requirements and verification

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	Test — lifecycle and stock
FR-03	number orders and shipments and show owner tracking (the staff member or driver currently holding each order)	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	Test — inventory and alerts
FR-05	record event notifications and allow reading	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	Test — aggregates and export
FR-07	authenticate and enforce role and object access	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	Test — assignment and delivery

NFR-01 performance (API p95 ≤2 s, first contentful paint ≤3 s); NFR-02 integration health (green build, test and browser runs recorded per increment; history in §6); NFR-03 least privilege and data minimisation — role gates on every mutating endpoint; staff order responses omit customer identity and delivery address (privacy tests); NFR-04 scripted operation effort — every core operation completes in at most three actions (walkthrough, §8); NFR-05 process quality — CI-verified increments (workflow).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	Increment register	scheduling plan
RK-02 contract rework	Transition tests pin the FR-02 contract (NFR-05)	transition tests
RK-03 test-week capacity	Priority register	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	Seed regression test locks FR-01/04/06	seed regression test
RK-06 assignments and test-week squeeze	Risk plan	assessment schedule
RK-07 stack unfamiliarity	Setup guide and CI lower ramp-up cost	setup guide
RK-08 report and video compete for time	Risk plan	report and video plan

4. System models

The four layers — Role-specific pages, HTTP routes and authentication, Order/inventory/driver business handlers, Prisma/SQLite persistence — each isolate one concern. Figures 1–5 answer, in order: which roles perform each function; which transitions and stock effects are valid; how items, orders, shipments and inventory logs relate; how confirmation commits or rolls back; how the four layers connect.

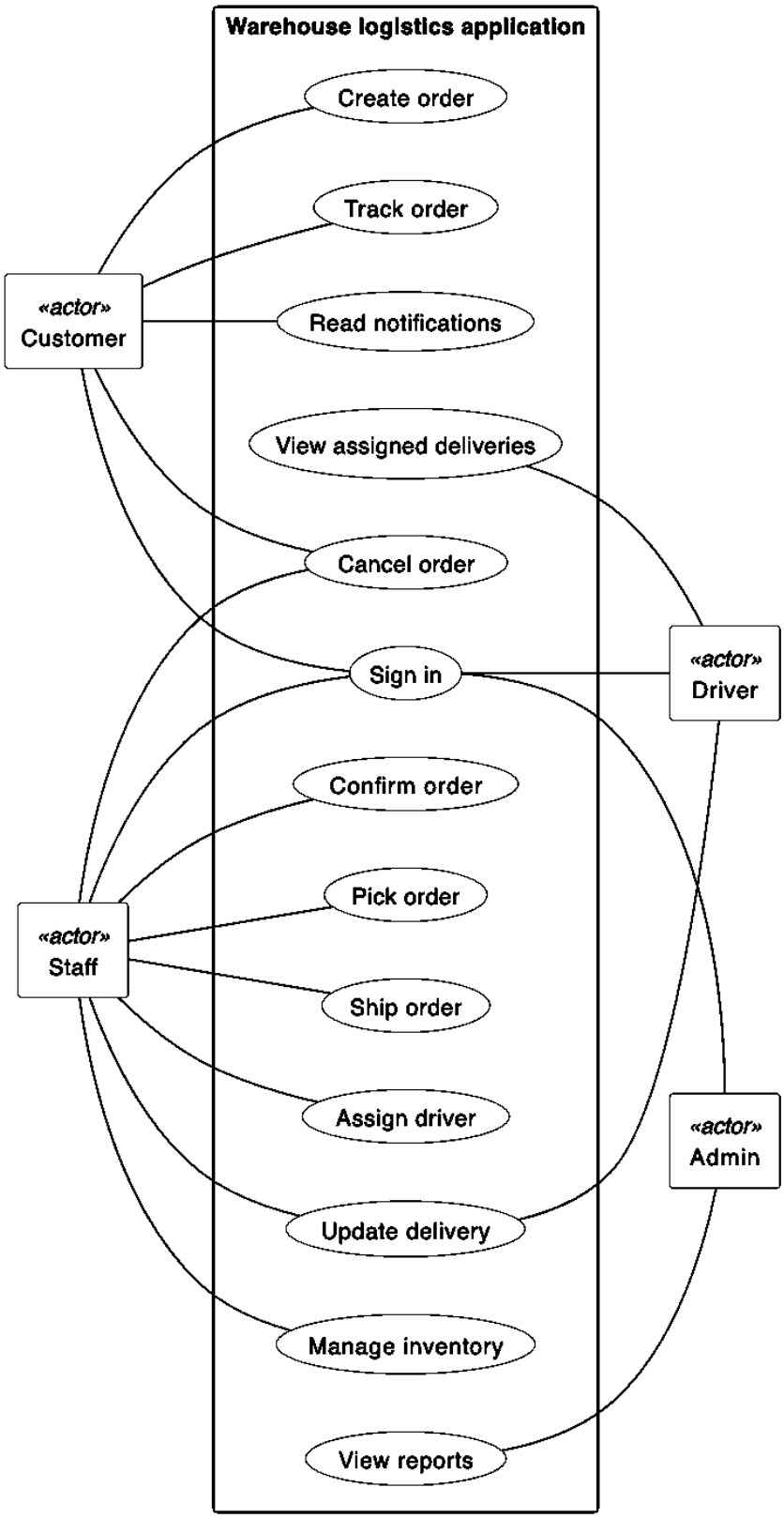


Figure 1. Role functions. Source.

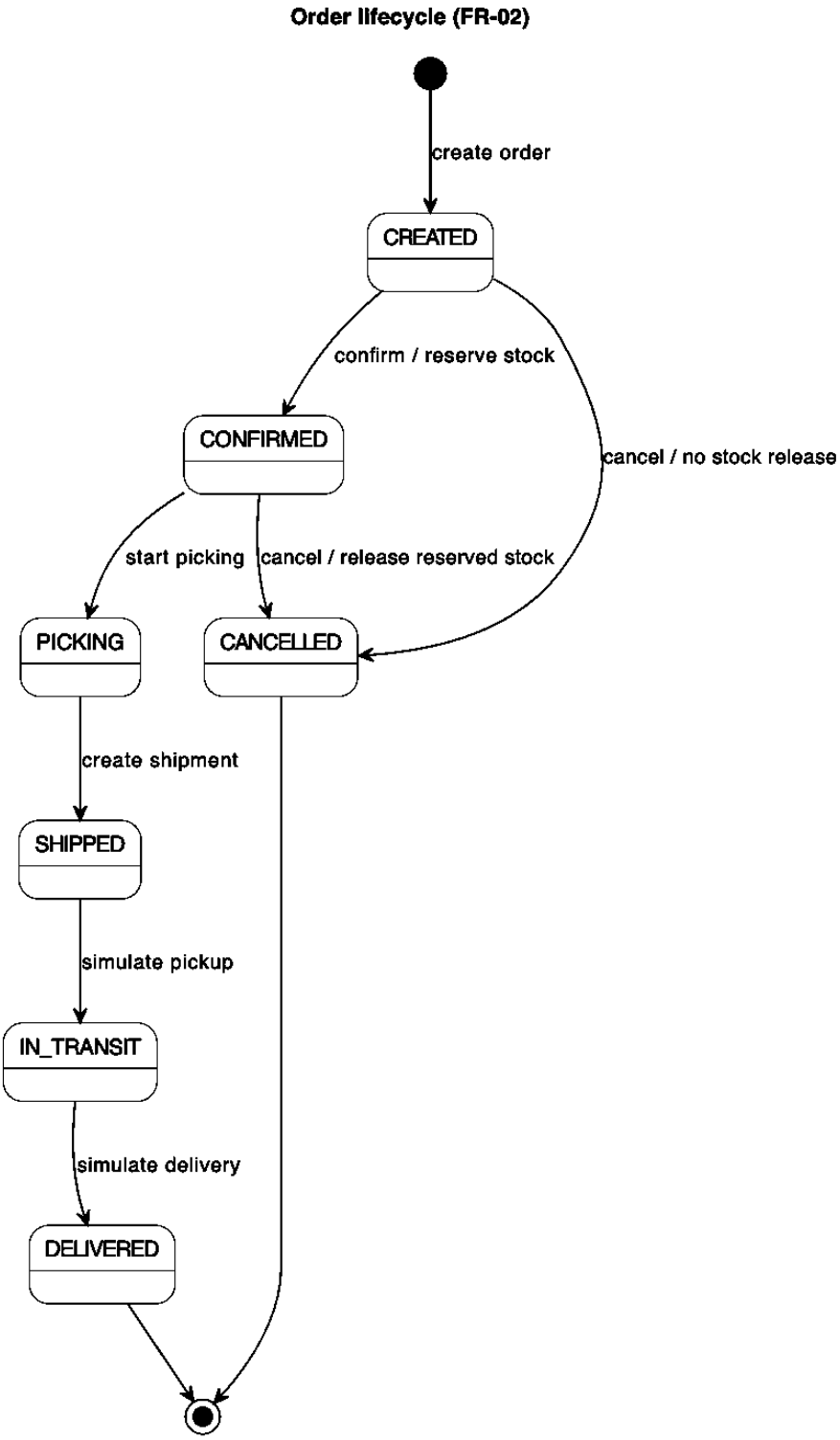


Figure 2. Order transitions. Source.

Each item references one product; each order requires at least one item; each order has 0..1 shipment; each product has 0..* inventory logs.

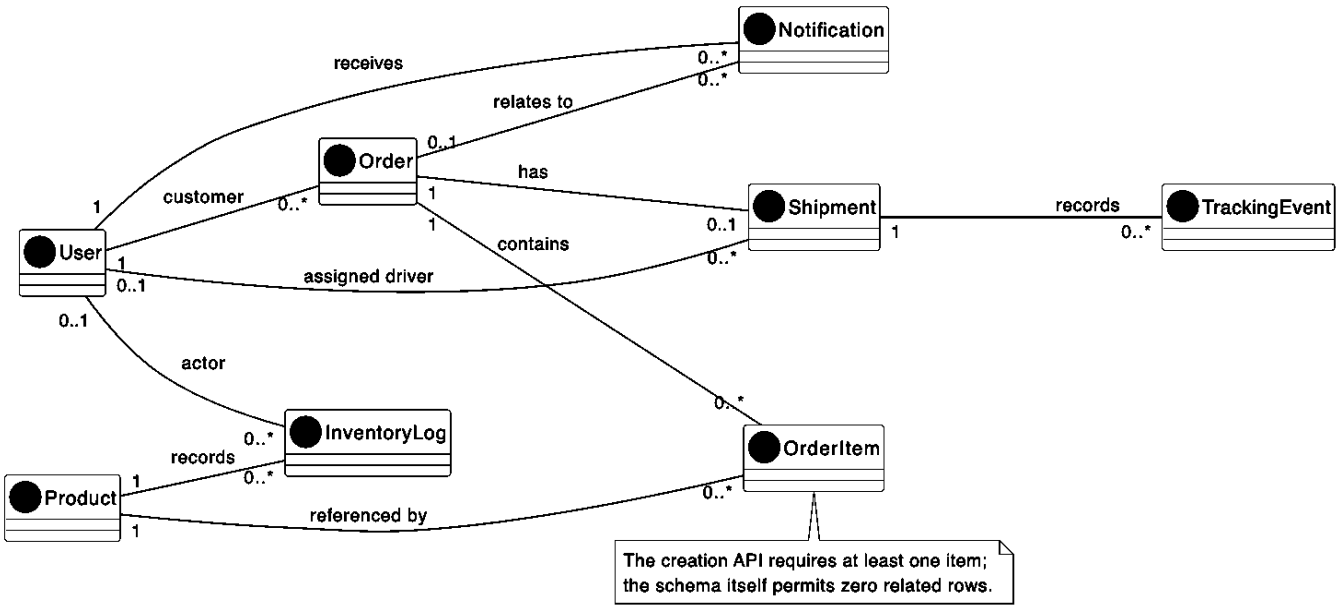


Figure 3. Data relationships. Source.

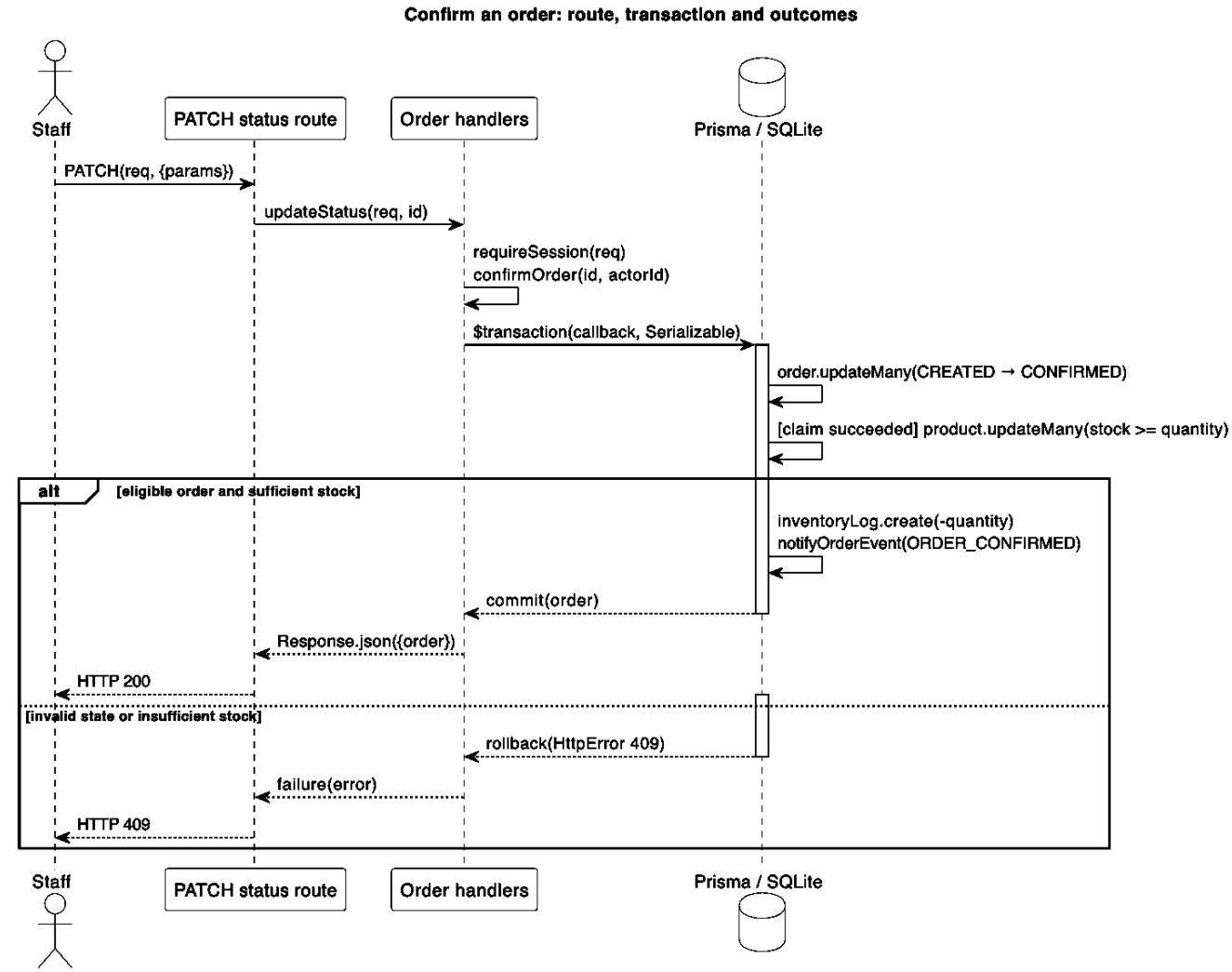


Figure 4. Confirmation transaction. Source.

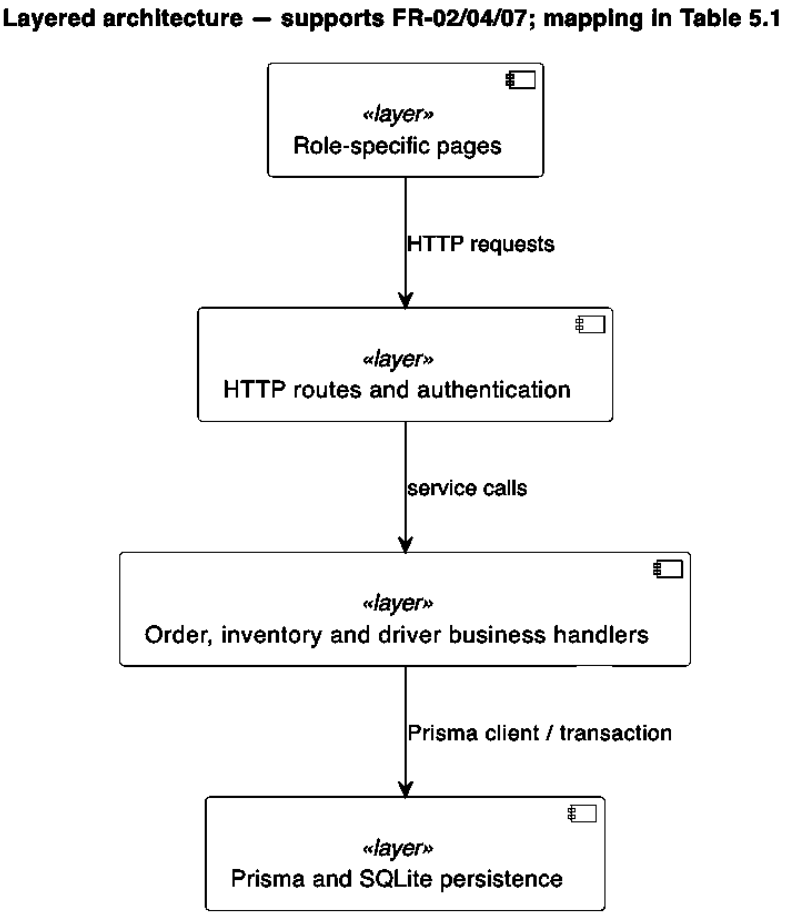


Figure 5. Four layers. Source.

5. Implementation

Stock is checked at creation and reserved at confirmation; the guards below and the confirmation transaction (order-api.ts:245–293) enforce both steps. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED (driver pickup and delivery); cancellation per the matrix below.

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: a transaction at the SERIALIZABLE isolation level writing state, stock, ledger and notifications	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 inventory ledger	order-api.ts:281, 398	Inventory

6. Verification and validation

Verification checks requirements against the implementation; validation exercises the delivered workflows through the interface (§8). The regression log and CI show 38 unit, 140 integration, 2 delivery checks (delivery checks), and 26 Playwright browser checks.

In a local-loopback measurement on 11 November 2026, API p95 was 3.2–13.6 ms across endpoints and first-contentful-paint p95 was 64–96 ms, against NFR-01 targets of at most 2 s and 3 s respectively. NFR-01 is met. NFR-02 is evidenced by the recorded green build, test and browser runs (observations).

7. Evolution

The 17 Oct 2026 test-led review — defects collected by exercising the running app — found a confirmation endpoint that accepted repeat submits and a missing stock ledger; the confirmation transaction and its retest followed (26–28 Oct). The 10–11 Nov documentation review corrected the diagrams and aligned traceability with the corrected models. Two integration-health regression tests added on 11 Nov brought the suite to 38.

8. Validation

The automated walkthrough exercises eleven core operations — create order; order details; confirm; pick; ship; assign; pickup; delivery; report filter; export report; selected-date export — each performed by its owning role, at most three actions each; every operation has expected outcomes asserted in the walkthrough script, and the Playwright 26 browser checks cover the eleven operations at one to three checks each. Results are logged with the walkthrough record.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagram bases diverged from the implemented schema	Models corrected	10–11 Nov documentation review	Diagra
Stock effects were recorded under FR-04	Moved to FR-02	17 Oct test-led review	Traceal
Trigger and outcome detail was thin	Expanded	10–11 Nov documentation review	Registe
The admin export took four scripted interactions	Re-tested	11 Nov automated walkthrough	Issue #
Integration-health checks had no regression test	Suite reached 38	11 Nov regression work	Test his
NFR criteria assumed a hosted deployment model	Set for local deployment	10–11 Nov documentation review	Require

10. Evidence

Traceability · Baseline record. All code, tests, evidence and logs are version-controlled in the repository.